

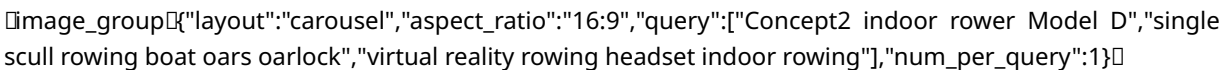
Evidence review: Ergometer-based versus boat-based inverse-kinematic mapping in rowing VR

Scope and terminology

This review focuses on whether there is published research that **directly compares** the effects of two different *inverse kinematics (IK) mapping paradigms* in **rowing virtual reality (VR)**:

- **Ergometer-based IK mapping:** avatar/body (and any visible “rowing apparatus”) is solved in a coordinate frame that is primarily anchored to the **indoor rowing machine** (handle/chain path and seat travel), i.e., the visuals match the machine’s kinematics.
- **Boat-based IK mapping:** avatar/body is solved in a coordinate frame anchored to a **virtual boat and oarlock geometry**, i.e., hand positions/orientations are mapped to oar angles and a boat-centric scene (often a single scull), even if the user physically rows on an ergometer.

The key question is not whether VR rowing exists (it does), nor whether IK is used (it is), but whether there is a **controlled, head-to-head comparison** that isolates **IK mapping choice** (erg-based vs boat-based) and measures its impact on outcomes such as technique, performance, exertion, presence, comfort, learning, or transfer.

The image group consists of two parts. The first part shows a Concept2 indoor rower Model D, which is a piece of exercise equipment used for rowing. The second part shows a virtual reality rowing headset, which is a device used for immersive virtual reality experiences.

Search approach and sources screened

This review uses a “snowball + targeted keyword” strategy:

- Targeted searches combined terms such as **rowing + virtual reality + ergometer + inverse kinematics + avatar + sculling + oar** and variants (“motion tracking”, “virtual coach”, “oar angle”, “boat-fixed coordinate system”). Several of the most relevant technical results were theses or workshop papers hosted by universities or author sites (e.g.,  “organization”, “University of Twente”, “university | enschede, nl”) repositories; author publication pages). ¹
- Citation chasing was done from recognised rowing-simulation platforms and papers (e.g., SPRINT, ETH’s rowing simulator work, and a set of rowing-VR exergame publications) into the underlying technical implementations. ²
- To reduce the risk of missing an “A/B mapping” study, I also screened broader **VR sport training** reviews that discuss gaps in transfer evidence and the state of VR training design. ³

A practical constraint in rowing-VR is that many papers evaluate **“VR vs non-VR”** (or different feedback/avatars/UI) rather than comparing alternative physical-to-virtual mappings, so the search explicitly looked for language indicating comparative mapping conditions (e.g., “two mapping strategies”, “boat coordinate vs machine coordinate”, “avatar mapping conditions”, “IK variants”).

What the literature does cover

Across the relevant body of work that is accessible and clearly described, the literature clusters into three overlapping themes. These themes **use IK and/or mapping**, but generally treat the mapping as an *implementation detail* rather than an experimental variable.

Ergometer-driven VR systems that use IK for avatar embodiment

A representative recent example implements a physical indoor rower and maps it to a **virtual single scull**, explicitly using IK to animate the avatar. In this system, participants row on a [entity["company","Concept2","indoor rowing manufacturer"]] machine while their motion is transferred into a sculling boat in VR; the authors track head and body points and animate the avatar with an IK package (Final IK by [entity["company","RootMotion","unity asset developer"]]) inside [entity["company","Unity","game engine company"]]. Importantly, the study's experimental manipulation is **avatar muscularity**, not the mapping paradigm; the "boat-based" rowing visualisation is constant. ⁴

Similarly, a line of student work around rowing technique feedback in VR uses an ergometer + HMD setup and discusses adding feedback and sensors; evaluations compare feedback conditions (e.g., adding a curve) rather than comparing mapping strategies. ⁵

Rowing "VR/VE training platforms" built around boat/oar models and technique feedback

The SPRINT programme (developed at [entity["organization","Sant'Anna School of Advanced Studies","university | pisa, it"]]) is influential for multimodal rowing training. Crucially for your question, SPRINT explicitly distinguishes between a **full mechanical platform** (two oars; supports sculling/sweep) and a **lightweight platform** based on a Concept2 ergometer, noting that the ergometer provides **limited kinematic features**. ⁶

SPRINT papers discuss capturing expert trajectories, calibrating user ranges, and delivering feedback (visual/vibrotactile/virtual humans). They also describe collecting data across different devices (SPRINT platform, Concept2 with motion capture, and instrumented boats). However, these comparisons are about **platform capability and training protocols**, not a controlled test of **erg-based vs boat-based IK mapping** while holding the physical set-up constant. ⁷

SPRINT's technique work includes detailed reconstruction of upper-body kinematics as kinematic chains in a defined world frame, based on motion-capture plus simulator data—again, technical reconstruction rather than comparative IK mapping conditions in VR. ⁸

High-fidelity rowing simulators with boat-fixed models and multimodal feedback

A separate stream (notably including [entity["organization","ETH Zurich","university | zurich, ch"]] affiliated work) focuses on physics-based rowing simulation, with explicit coordinate systems including **boat-fixed frames** and oar-fixed frames. ⁹

For example, a configurable 3D rowing model for indoor training defines multiple coordinate systems (including a boat-fixed frame) to model oar motion and force generation. ¹⁰
Other work examines how "virtual water" parameters influence motor learning in sophisticated rowing simulation contexts. ¹¹

A well-cited “virtual-to-real transfer” rowing study used a multimodal simulator with oar-water interaction depicted visually/auditorily and rendered haptically, aiming at realistic rowing interactions; its central comparison is about **learning/transfer**, not about alternative IK mapping paradigms for the same physical input device. ¹²

Earlier design work on rowing simulators integrated with VR describes building indoor systems that replicate aspects of outdoor rowing with multimodal cues; again, the mapping is part of building the system, not an A/B experimental factor. ¹³

Evidence relevant to “erg-based vs boat-based IK” as an experimental comparison

Across the sources above, **IK is present**, and both “ergometer-like” and “boat-like” kinematic representations are present, but I did not find research that does the following simultaneously:

1. Uses *the same physical rowing input* (e.g., same ergometer, same sensors, same pace/force targets), and
2. Implements *two different IK mapping paradigms* (ergometer-based vs boat-based) as the **independent variable**, and
3. Reports a controlled evaluation attributing outcome differences to the mapping choice.

Instead, the controlled comparisons found in the rowing-VR neighbourhood tend to be of different types:

- **VR vs non-VR** comparisons (e.g., a pilot study placing a rower on a stationary rowing machine in a VR environment and reporting improved workout performance/experience vs standard rowing). ¹⁴
- **Feedback design comparisons**, such as showing additional performance curves/visualisations; these change feedback availability, not the underlying embodiment mapping paradigm. ¹⁵
- **Avatar property comparisons** (e.g., muscular vs non-muscular avatar), where the mapping and IK approach are fixed. ⁴
- **Simulator parameter/training schedule comparisons** (e.g., manipulating simulated water density/difficulty or other task parameters), again not changing erg-vs-boat IK mapping for the same physical control. ¹¹
- **Social/companion presence manipulations** during rowing exercise in VR, focusing on motivational/affective outcomes rather than embodiment mapping. ¹⁶

There is also adjacent non-VR biomechanics work comparing on-water rowing and ergometer rowing kinematics, indicating that the two contexts can differ in meaningful ways. This strengthens the rationale for your question (because mapping could plausibly matter), but it does not constitute a VR IK mapping comparison. ¹⁷

Bottom line on whether the gap is real

Based on the material located and inspected, **there does not appear to be a peer-reviewed, clearly documented online study that directly compares “ergometer-based IK mapping” versus “boat-based IK mapping” in rowing VR as the central experimental factor.**

What *does* exist is a patchwork where:

- Researchers **implement** a mapping (sometimes explicitly boat-based sculling visuals driven by ergometer trackers and IK) and then study something else (avatars, UI, feedback, social presence). ¹⁸
- Simulator researchers build systems whose kinematics are inherently boat/oar-centric (or split into “full oars” vs “ergometer” platforms) but evaluate training protocols, feedback, or learning/transfer rather than mapping variants. ¹⁹
- Reviews of VR sports training note that transfer evidence is limited/inconsistent and that design issues (including body visualisation) remain open—suggesting plenty of room for more controlled design-factor studies, though they do not single out “erg vs boat IK mapping” as a studied variable. ³

So, the evidence supports the conclusion that the **specific comparison you care about is either unstudied or not readily findable in standard public/academic channels**, even though the underlying ingredients (rowing VR, ergometer integration, embodiment via IK, boat-fixed modelling) are all present in the broader literature. ²⁰

Implications for designing the missing comparison study

If you decide to fill this gap (or evaluate it internally), the closest “research-shaped” experiment would treat IK mapping paradigm as the only major difference:

- Keep the **physical device** constant (same ergometer, same drag/resistance, same sensor suite).
- Keep the **feedback** constant (same stroke rate targets, same performance HUD availability, same audio).
- Implement two conditions:
 - **Erg-based visual/IK**: the avatar rows a virtual ergometer; hands follow erg handle path; body IK solves in machine frame.
 - **Boat-based visual/IK**: the avatar rows a virtual scull/boat; hands are constrained by oarlock geometry; IK solves to meet boat-defined handle constraints (oar rotation about riggers), and boat movement is driven by the stroke.
- Measure outcomes at three levels:
 - **Immediate experience**: presence, comfort/VR sickness, perceived realism.
 - **Performance and technique proxies**: stroke timing consistency, trunk/arm joint kinematics (if tracked), force/pace stability.
 - **Retention/transfer** if training: performance/technique in a non-VR follow-up (and/or on-water if feasible), consistent with the skill-transfer framing used in rowing simulator work. ²¹

This design aligns with how existing systems already calibrate user size/range and represent rowing skill digitally (e.g., SPRINT’s calibration and trajectory representations, and sculling-in-VR setups that align virtual seat travel and oar rotation geometry). ²²

¹ ⁵ ¹⁵ https://essay.utwente.nl/fileshare/file/87776/Westra_BA_EEMCS.pdf
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